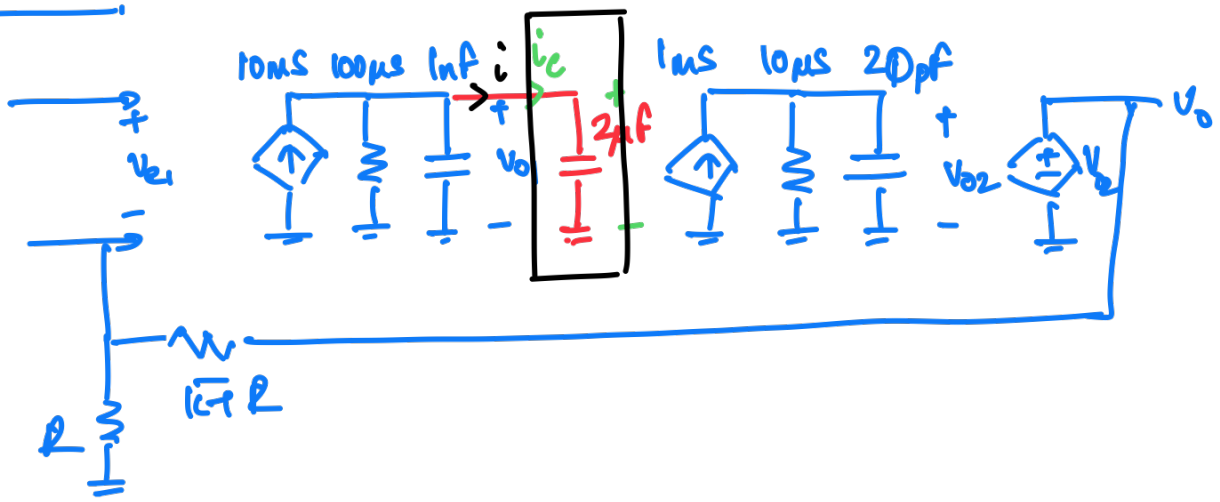
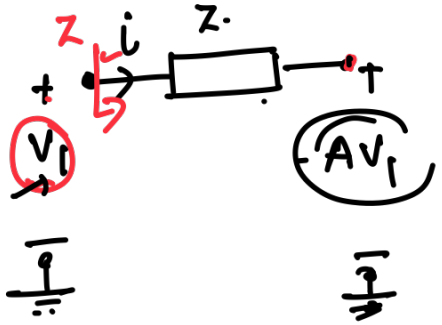
Lecture #16



$i_c = V_{o1}\, sC_x$

$$\underline{I_c = i} = V_{o1}\, sC_x = \frac{V_{o1}}{Z_1} = \frac{\alpha V_{o1}}{\beta Z_1}$$



$$i = \frac{V_i - (-AV_1)}{Z_1} = \frac{V_1}{Z_1/(1+A)}$$

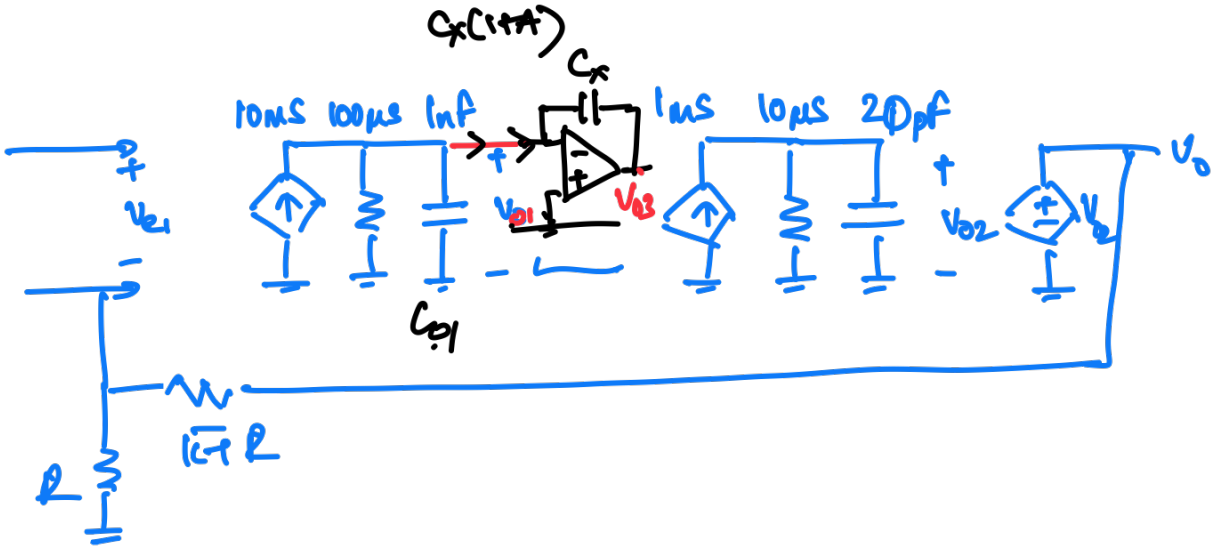
$$\frac{V_1}{i} = \frac{Z_1}{(1+A)}$$

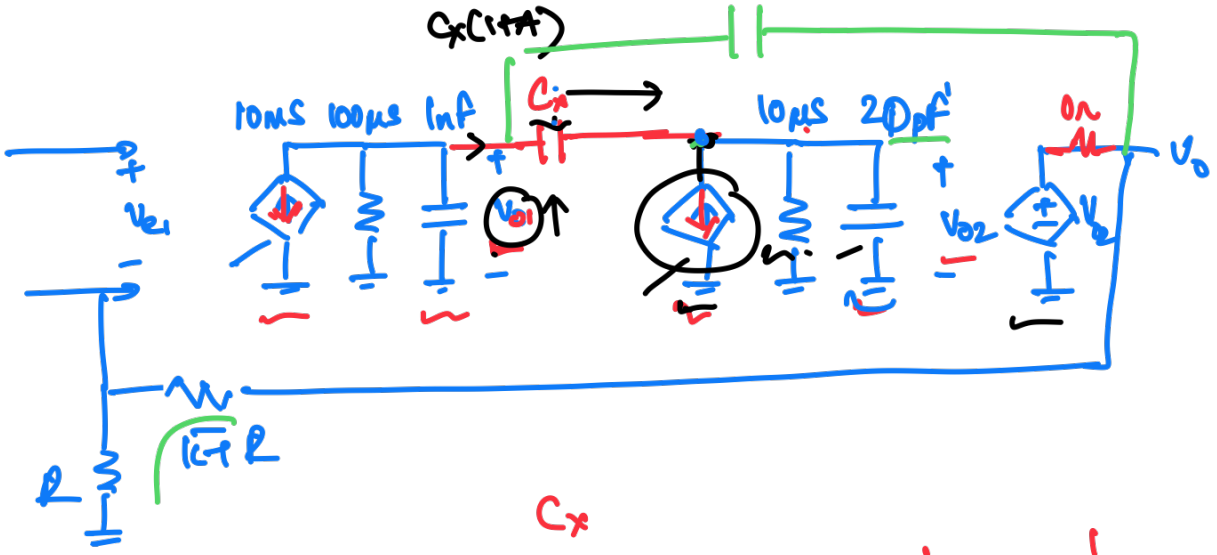
if $Z_1 = \frac{1}{sC_x}$

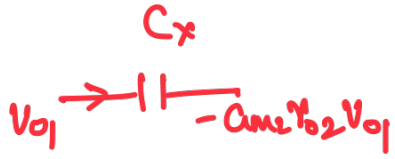
$$\frac{V_1}{i} = \frac{1}{sC_x(1+A)} = \frac{1}{sC_y}$$

$$C_y = C_x(1+A)$$

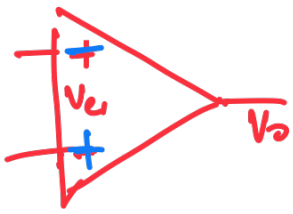
"Miller Multiplication"



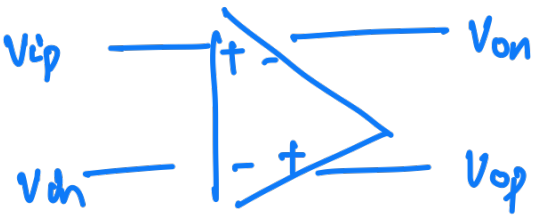




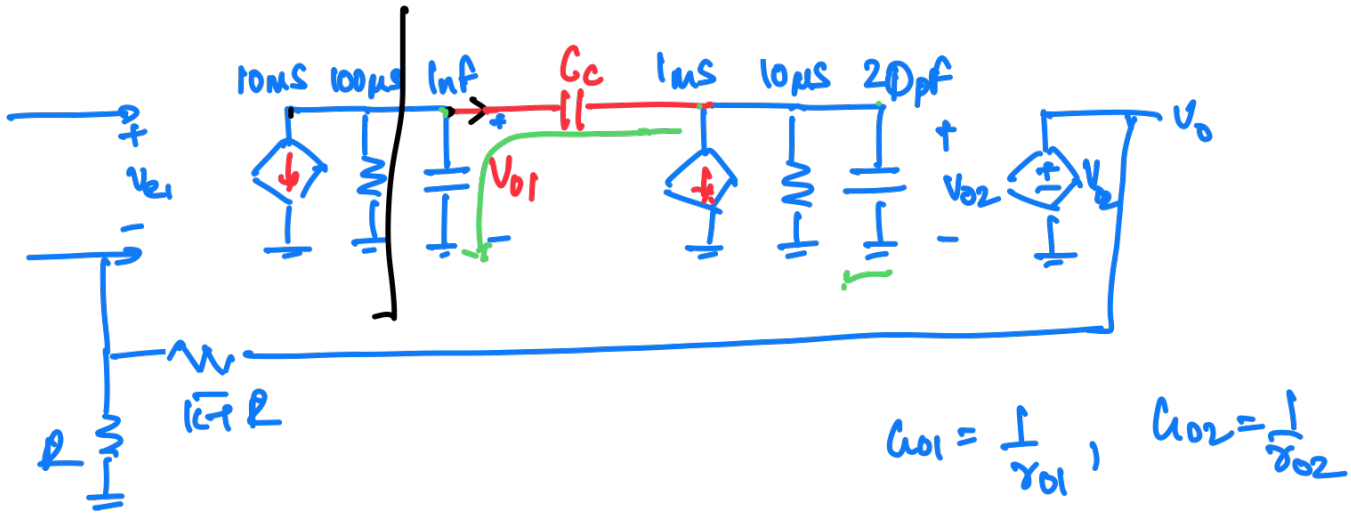
$$i = \frac{1}{sC_x(1+A_{v2}R_{o2})} V_{o1}$$



$$V_o = AV_e$$



$$V_{op} - V_{on} = A(V_{ip} - V_n)$$



$G_{o1} = \frac{1}{r_{o1}}$, $G_{o2} = \frac{1}{r_{o2}}$

$$G_{m1}V_i + \frac{V_{01}}{r_{01}} + V_{01}sC_{01} + (V_{01} - V_{02})sC_c = 0$$

$$G_{m1}V_i + V_{01}(G_{o1} + sC_{01} + sC_c) - V_{02}sC_c = 0 \quad -(1)$$

$$G_{m2}V_{01} + V_{02}(G_{o2} + sC_{02}) + (V_{02} - V_{01})sC_c = 0$$

$$(G_{m2} - sC_c)V_{01} + V_{02}(G_{o2} + sC_{02} + sC_c) = 0 \quad -(2)$$

$$G_{m1}V_i - V_{02}sC_c + (G_{o1} + sC_{01} + sC_c) \times -\frac{V_{02}(G_{o2} + sC_{02} + sC_c)}{(G_{m2} - sC_c)}$$

$$+ V_{02}\left[\frac{sC_cG_{m2} - s^2C_c^2 + (G_{o1} + sC_{01} + sC_c)(G_{o2} + sC_{02} + sC_c)}{(G_{m2} - sC_c)}\right]$$

$$= G_{m1}V_i$$

$$\frac{V_{02}}{V_i} = \frac{G_{m1}(G_{m2} - sC_c)}{G_{o1}G_{o2} + s(C_{02} + C_c)G_{o1} + sC_{01}G_{o2} + s^2C_{01}C_c + sC_cG_{o2} + s^2C_{01}C_{02} + s^2C_cC_{02} + sC_cG_{m2}}$$

$$LG = \frac{1}{K} \frac{G_{m1}G_{m2}\left(1 - sC_c/G_{m2}\right)}{G_{o1}G_{o2} + s\left((G_{o2}+C_c)G_{o1} + (G_{o1}+C_c)G_{o2} + C_cG_{m2}\right) + s^2\left(C_{o1}C_{o2} + C_cC_{o2} + C_cC_{o1}\right)}$$

$$= \frac{1}{K} \frac{G_{m1}r_{o1}\,G_{m2}r_{o2}\left(1 - sC_c/G_{m2}\right)}{1 + s\left((C_{o2}+C_c)r_{o2} + (C_{o1}+C_c)r_{o1} + C_cG_{m2}r_{o1}r_{o2}\right) + s^2\left(C_{o1}C_{o2} + C_cC_{o2} + C_cC_{o1}\right)r_{o1}r_{o2}}$$

$$= \frac{1}{K} \frac{G_{m1}r_{o1}\,G_{m2}r_{o2}\left(1 - sC_c/G_{m2}\right)}{1 + s\left((C_{o2}+C_c)r_{o2} + \underbrace{(C_{o1}+C_c + C_cG_{m2}r_{o2})}_{C_c(1+G_{m2}r_{o2})}r_{o1}\right) + s^2\left(C_{o1}C_{o2} + C_cC_{o2} + C_cC_{o1}\right)r_{o1}r_{o2}}$$

$$\left\{C_{o1} + C_c\left(1 + G_{m2}r_{o2}\right)\right\}r_{o1}$$

$$D(s) = s^2\left(C_{o1}(C_{o2}+C_c) + C_cC_{o2}\right)r_{o1}r_{o2} + s\left((C_{o2}+C_c)r_{o2} + \left(C_{o1} + C_c(1+G_{m2}r_{o2})\right)r_{o1}\right) + 1 = ax^2 + bx + c$$

(higher freq. pole)

$$p_2' = -\frac{\left((C_{o2}+C_c)r_{o2} + \left(C_{o1} + C_c(1+G_{m2}r_{o2})\right)r_{o1}\right)}{\left(C_{o1}(C_{o2}+C_c) + C_cC_{o2}\right)r_{o1}r_{o2}}$$

$$p_1' = -\frac{1}{\left((C_{o2}+C_c)\,r_{o2} + \left(C_{o1} + C_c\left(1+g_{m2}r_{o2}\right)\right)r_{o1}\right)}$$

(Dominant pole)

$C_c\left(1+g_{m2}r_{o2}\right) \gg C_{o1}$

$$p_1 = -\frac{1}{r_{o1}C_{o1}} > p_1'$$

$$p_2 = -\frac{1}{r_{o2}C_{o2}}$$

$$p_2' = -\frac{\left((C_{o2}+C_c)\,r_{o2} + \left(C_{o1} + C_c\left(1+g_{m2}r_{o2}\right)\right)r_{o1}\right)}{\left(C_{o1}\left(C_{o2}+C_c\right) + C_cC_{o2}\right)r_{o1}r_{o2}}$$

$$= -\frac{}{\left(C_{o1}+C_c\right)\left(C_{o2} + \frac{C_{o1}C_c}{C_{o1}+C_c}\right)r_{o1}r_{o2}}$$

$\frac{C_{o1}C_c}{C_{o1}+C_c} \approx C_{o1}$